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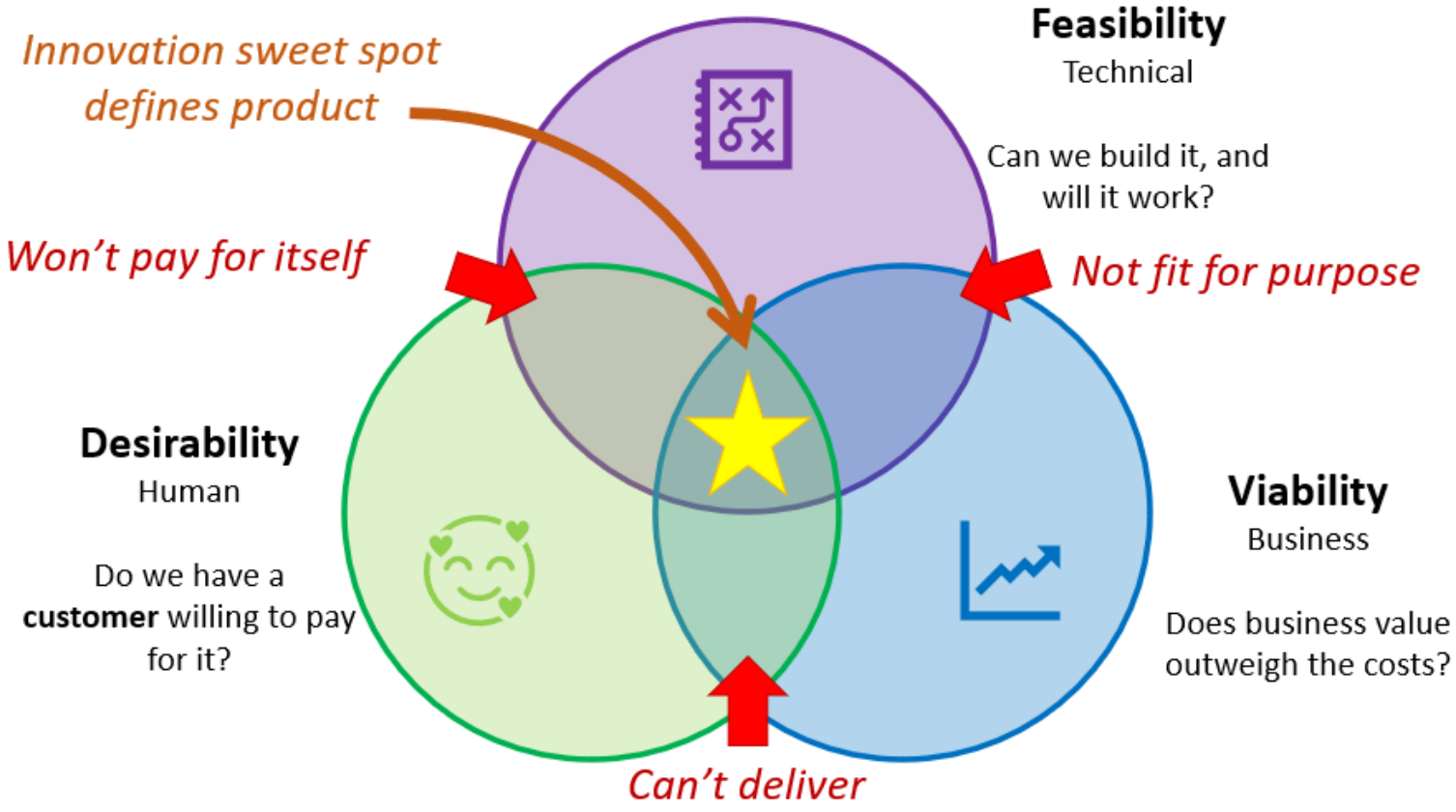
ÉCOLE POLYTECHNIQUE DE LOUVAIN

LEPL2022 – Health Innovation Classes
Regulations and guidelines in health – Pierre Geens

April 2nd, 2026

Desirability
Feasibility
Viability



Source: https://www.linkedin.com/pulse/desirability-feasibility-viability-successful-data-science-dods-phd/

Impacts on Innovation

WHAT ABOUT REGULATORY?

DESIRABILITY

DOES THE LAWMAKER PUBLICLY EXPRESS SPECIFIC OPINIONS/DESIRES IN FAVOUR OR AGAINST THE CONCEPT?

FEASIBILITY

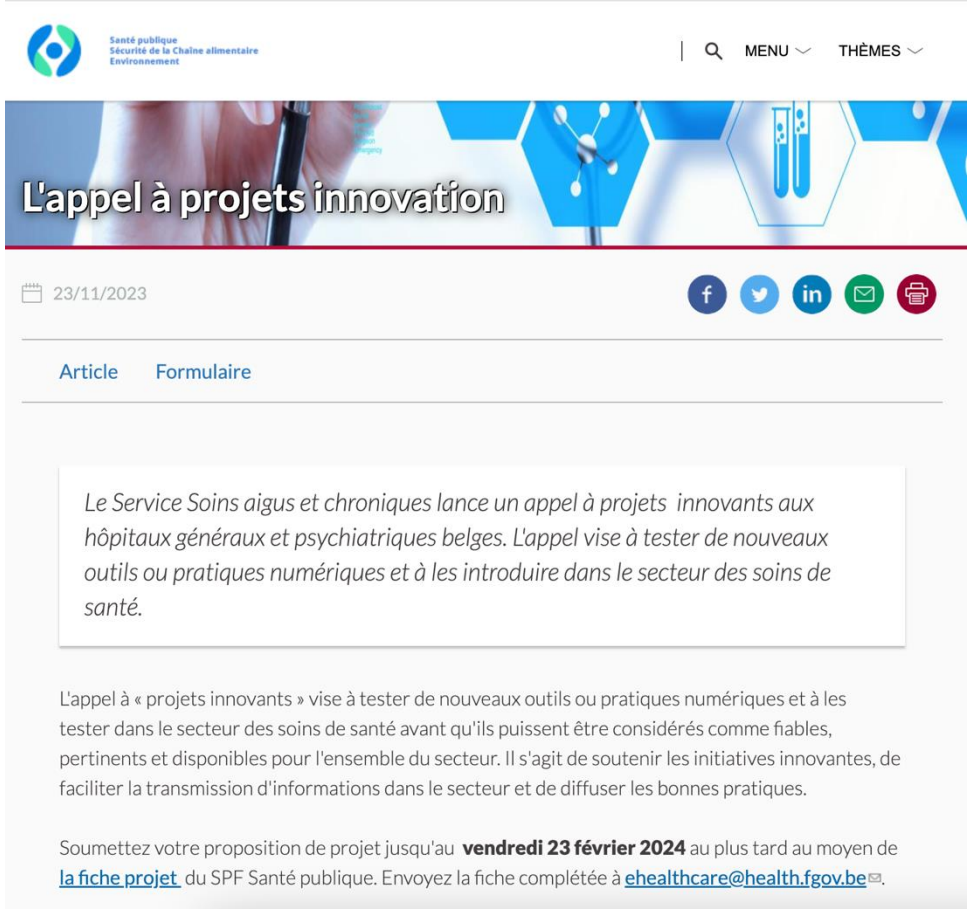
IS IT EVEN PERMITTED?

DO WE HAVE ADEQUATE RESOURCES TO COMPLY?

VIABILITY

IS THE COST OF COMPLIANCE BEARABLE vs REVENUES?

DESIRABILITY



Santé publique
Sécurité de la Chaîne alimentaire
Environnement

MENU THÈMES

L'appel à projets innovation

23/11/2023

Article Formulaire

Le Service Soins aigus et chroniques lance un appel à projets innovants aux hôpitaux généraux et psychiatriques belges. L'appel vise à tester de nouveaux outils ou pratiques numériques et à les introduire dans le secteur des soins de santé.

L'appel à « projets innovants » vise à tester de nouveaux outils ou pratiques numériques et à les tester dans le secteur des soins de santé avant qu'ils puissent être considérés comme fiables, pertinents et disponibles pour l'ensemble du secteur. Il s'agit de soutenir les initiatives innovantes, de faciliter la transmission d'informations dans le secteur et de diffuser les bonnes pratiques.

Soumettez votre proposition de projet jusqu'au **vendredi 23 février 2024** au plus tard au moyen de la fiche projet du SPF Santé publique. Envoyez la fiche complétée à ehealthcare@health.fgov.be.

IDEA CONSULT thinking ahead

Renouvellement de la Stratégie de Spécialisation Intelligente de la Wallonie (S3) – Fiche Domaine d'Innovation Stratégique
« Innovations pour une santé renforcée'

30 Octobre 2020

Personne de contact:

Dr. Vincent Duchêne
Senior Expert & Managing Partner

T: +32/472 52 28 28
vincent.duchene@ideaconsult.be

40, Rue Joseph II - B1
1000 Bruxelles

T: +32 2 282 17 10
info@ideaconsult.be

Pour

Service Public de Wallonie Direction de la Politique Économique
1, Place de la Wallonie,
5100 Namur

Cahier spécial des charges nr 06.04.01-20-0154

https://www.health.belgium.be/fr/news/lappel-projets-data-capabilities#article
https://s3.wallonie.be/files/Documents/Doc%20DIS/DIS%20Sant%c3%a9.pdf

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FEASIBILITY

Portable home-use X-ray for self assessment of dental health

Value Propositions

- Increased Accessibility
  - Improved Dental Care Access:
  - Convenience
- Early Detection and Prevention
  - Proactive Monitoring
  - Prevention of Major Problems:
- Cost Savings
  - Reduction in Clinical Visits
  - Lower Overall Expenses
- Technological Advancement
  - Improved Miniaturization
  - AI Integration
- Empowering Patients
  - Promoting Self-Care
  - Educational Tool
- Pandemic and Public Health Benefits
  - Reduction of In-Person Visits
  - Support During Public Health Crises
- Etc…

Customer Re

Channels

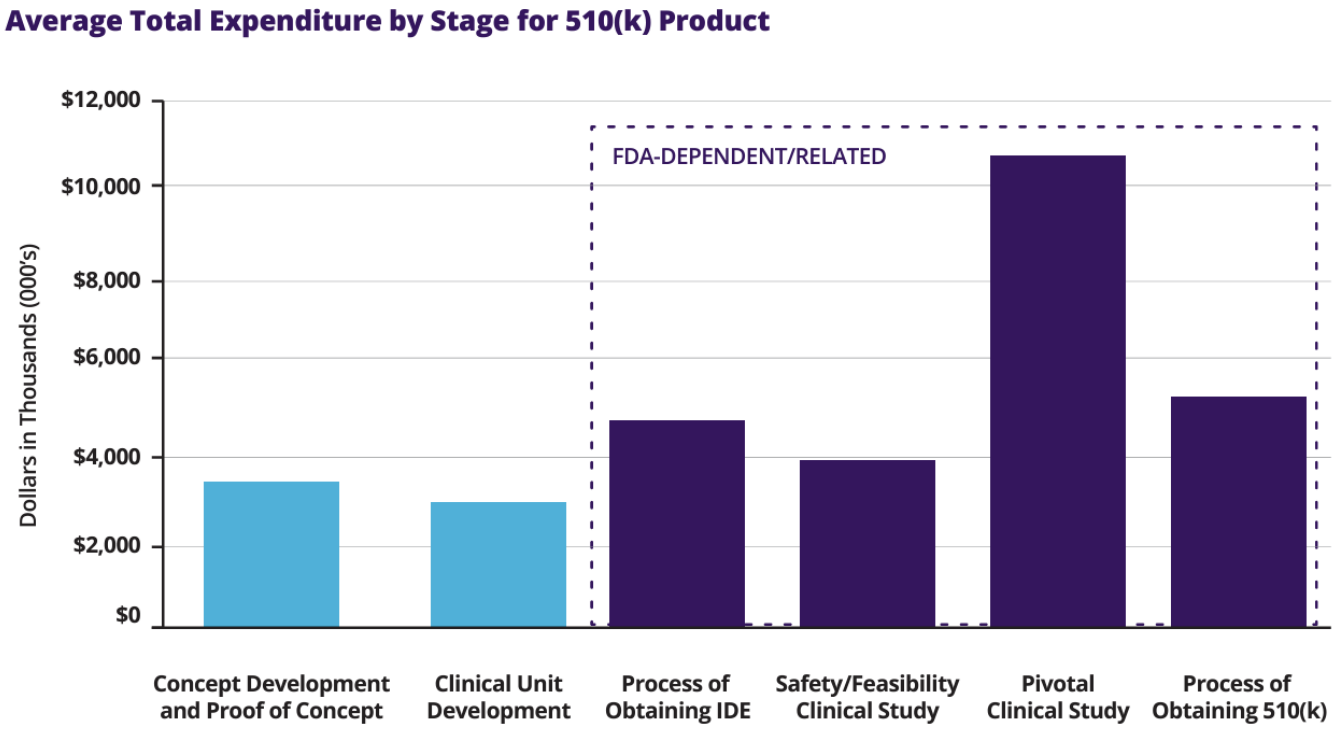
FEASIBILITY

Portable home-use X-ray for self assessment of dental health

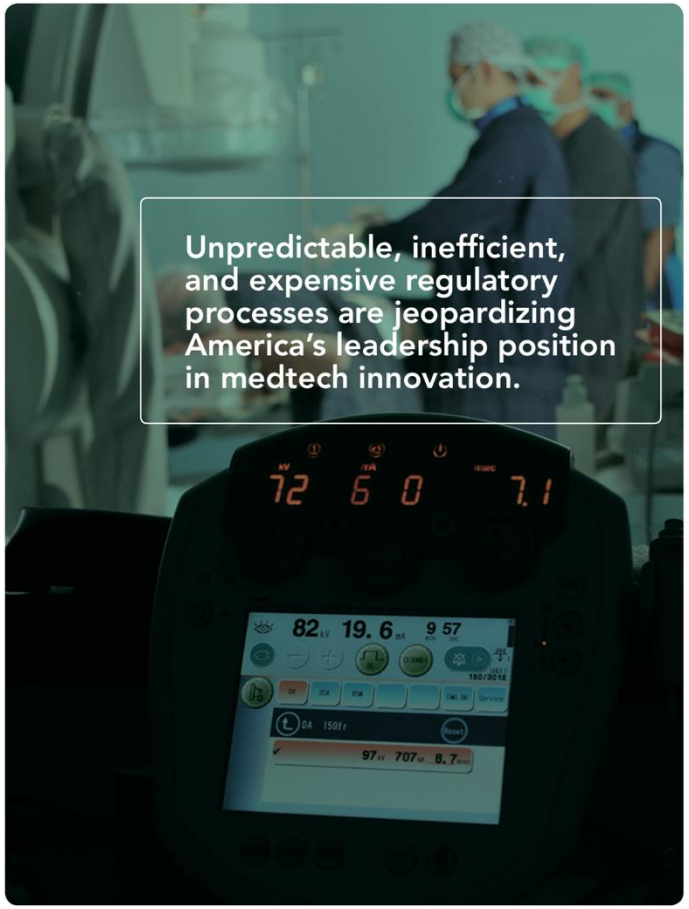
EURATOM Directive 2013/59, national regulations, ethical principles,…

- These laws require the use of X-rays to be justified by a doctor and carried out under medical supervision in a controlled environment.
- According to international standards, exposure to ionising radiation must be justified by a medical benefit that outweighs the risks.
- Such a device would not be able to obtain the CE mark required to be marketed in the EU, as it would not meet the essential requirements
- The absence of medical supervision poses major health risks to users, including repeated and unsupervised exposure to high doses of X-rays, and misinterpretation of results.

VIABILITY



https://starfishmedical.com/assets/StarFish-Whitepaper-Cost-to-Develop-Medical-Devices-July-2020.pdf



https://www.medtecheurope.org/resource-library/fda-impact-on-us-medical-technology-innovation

WHY DO YOU NEED TO ANTICIPATE?

Ideation

- Assess regulatory feasibility
- Encourage ethical practices
- Enhance safety and efficacy

Development

- Ensure compliance
- Reduce costs
- Support sustainable development

Market access

- Accelerate time-to-market
- Build stakeholder trust
- Facilitate global reach

Goals and Structure

DIFFERENT GOALS

Patient-centered protections

Protect patients rights and safety

- Regulations establish standards to safeguard patients' rights, ensuring they receive safe, ethical, and high-quality care. This includes maintaining confidentiality, obtaining informed consent, and providing accurate information about treatments.

Protect healthcare data and privacy

- Guidelines like the Health Insurance Portability and Accountability Act (HIPAA) in the U.S. ensure the confidentiality and security of patient information, fostering trust in the healthcare system.

Public Health and Safety

Maintaining Public Health and Safety

- Implement measures to prevent and control the spread of diseases through vaccination mandates, infection control protocols, and reporting requirements.

Promoting Ethical Research and Innovation

- Health regulations support ethical research and innovation by ensuring safety, protecting participants, enforcing standards, promoting transparency, guiding responsible development benefits society and builds public trust.

DIFFERENT GOALS

Standardization and Quality Assurance

Standardize Care Delivery

- Promote uniformity in healthcare practices, reducing variability in patient care and ensuring evidence-based treatments

Ensuring Healthcare Provider Competence

- Oversee the licensing and continuous education of healthcare professionals to maintain high standards of care.

System Efficiency and Accountability

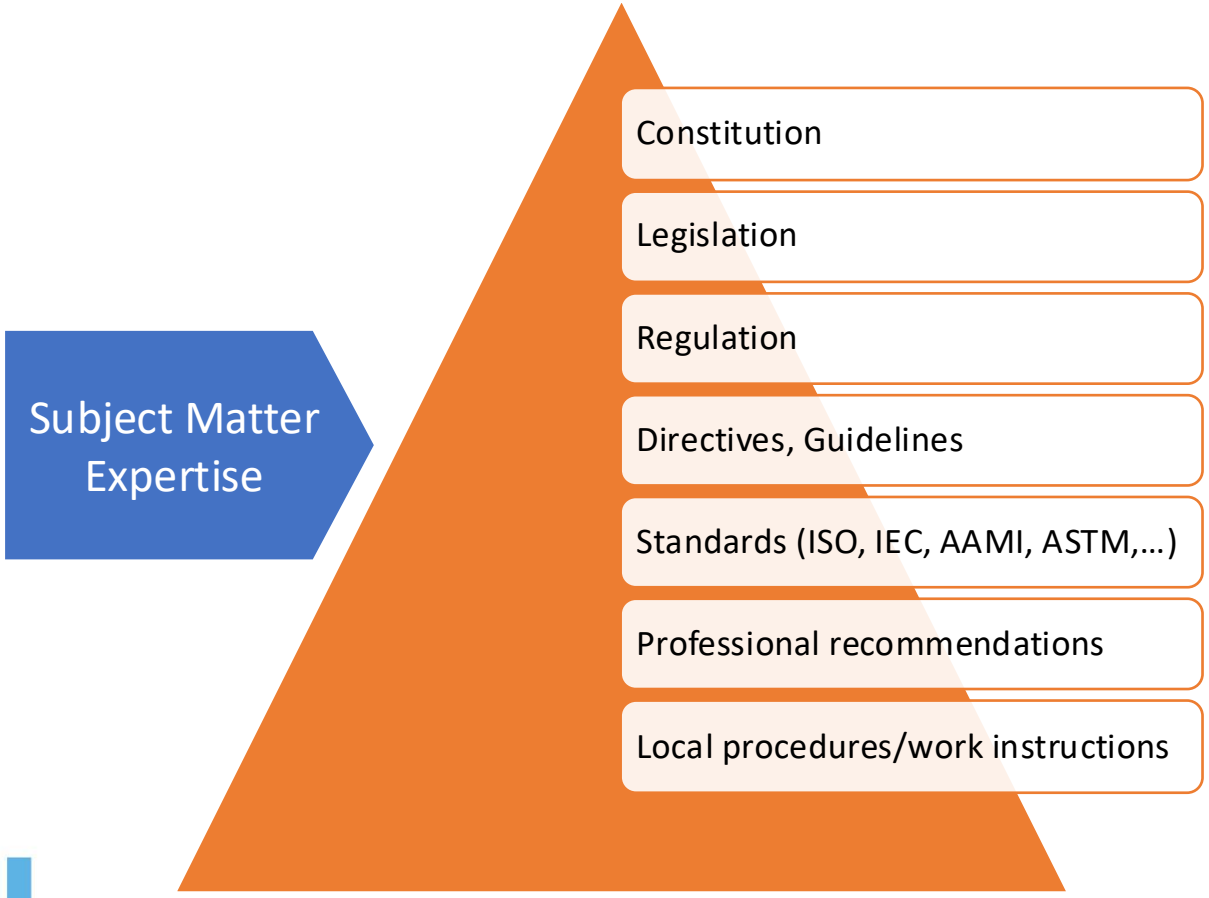
Facilitating System Efficiency and Accountability

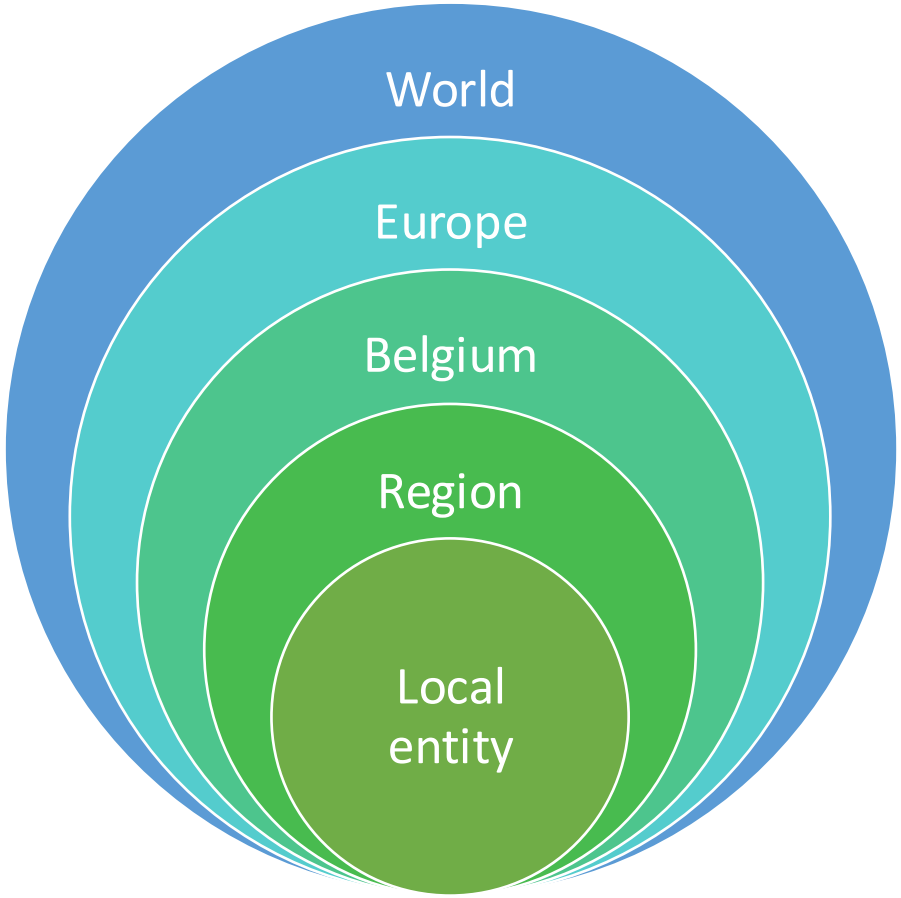
- Set operational standards and performance benchmarks to enhance the efficiency of healthcare delivery and hold institutions accountable.

Ensuring Financial Integrity and Fairness

- Address billing practices, insurance coverage, and reimbursement rates to prevent fraud and promote equitable access to healthcare services.

DIFFERENT LEVELS





EXAMPLE 1: MEDICAL SOFTWARE CYBERSECURITY

GHTF/SG1/N78:201
Principles of Conformity Assessment for Medical Devices

EU
MDR 2017/745
Medical Devices Regulation

US
CFR Title 21 – Chapter I – Subpart H
Medical Devices

Canada
SOR/98-282
Medical Device Regulation

Etc…

la loi du 22 décembre 2020 relative aux dispositifs médicaux et ses arrêtés d'exécution

MDCG 2019-16 rev.1
Guidance on cybersecurity for medical devices

MDCG 2023-4
Medical Device Software (MDSW) – Hardware combinations

Etc…

IEC 62304

ISO 14971

AAMI TIR57:2016

etc…

EXAMPLE 2: IONISING RADIATION IN HEALTH

International Atomic Energy Agency
IAEA Human Health Reports No5.
Status of Computed Tomography Dosimetry for Wide Cone Beam Scanners

European Union. Council Directive 2013/59/Euratom of 5 December 2013 laying down basic safety standards for protection against the dangers arising from exposure to ionising radiation

European Union. Radiotherapy Directorate-General for Energy Directorate D — Nuclear Safety & Fuel Cycle Unit D4. Criteria for Acceptability of Medical Radiological Equipment used in Diagnostic Radiology, Nuclear Medicine and— Radiation Protection 2012

Loi du 10 juillet 2008 Loi coordonnée sur les hôpitaux et autres établissements de soins

Arrêté royal du 20 juillet 2001 portant règlement général de la protection de la population, des travailleurs et de l'environnement contre le danger des rayonnements ionisants

Arrêté royal du 18 février 2014 fixant les critères minimaux d'acceptabilité pour les appareils de scanographie (scanners CT) utilisés à des fins d'imagerie médicale

Arrêté royal du 25 avril 2014 portant la liste de l'appareillage médical lourd au sens de l'article 52 de la loi coordonnée sur les hôpitaux et autres établissements de soins

Arrêté royal du 19 janvier 2016 déterminant les règles suivant lesquelles les données relatives à l'appareillage médical lourd sont communiquées au Ministre qui a la Santé publique dans ses attribution

Protocole d'accord relatif à l'imagerie médicale

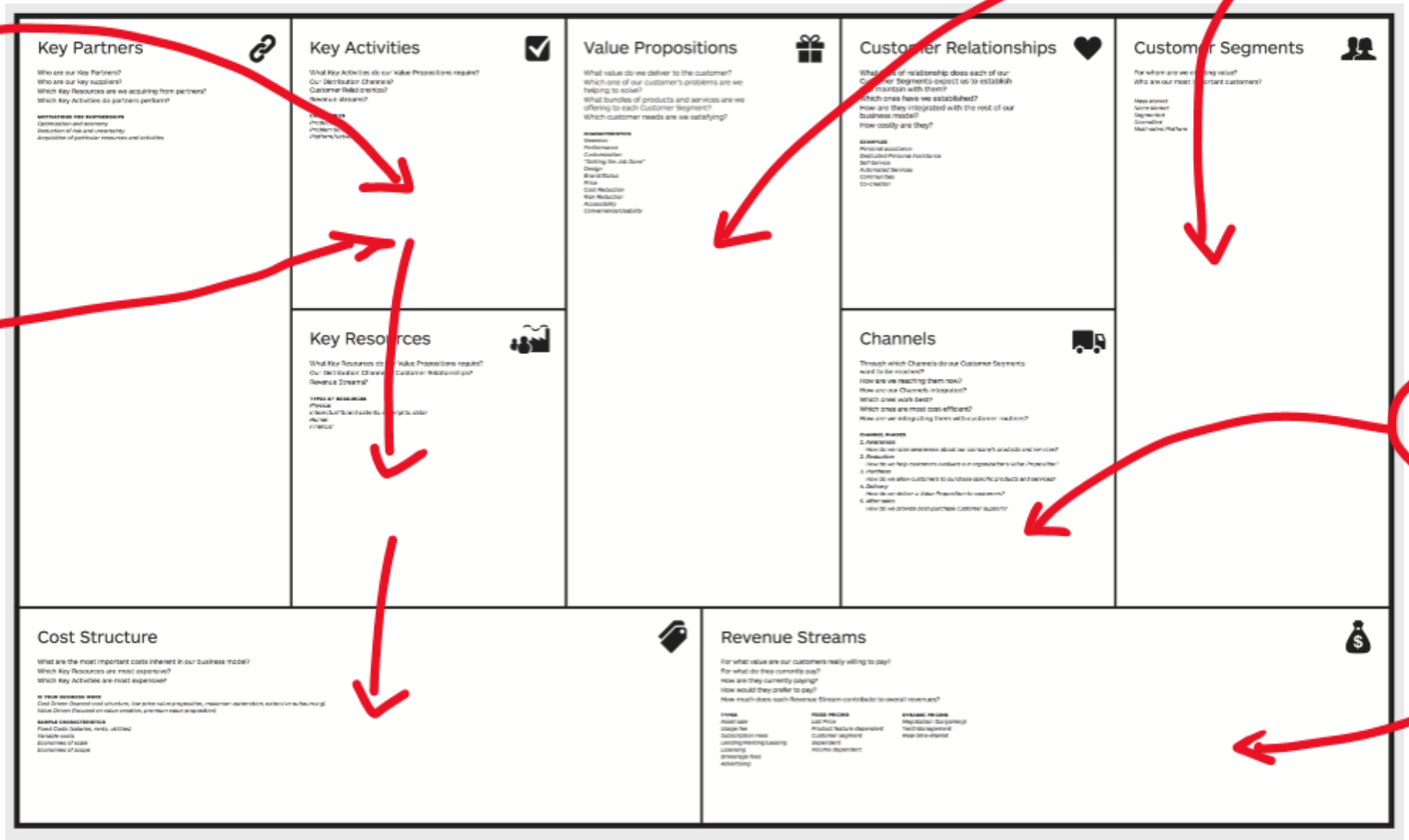
Etc…

WHAT'S THE IMPACT?

Concept itself

Build compliance

Product design

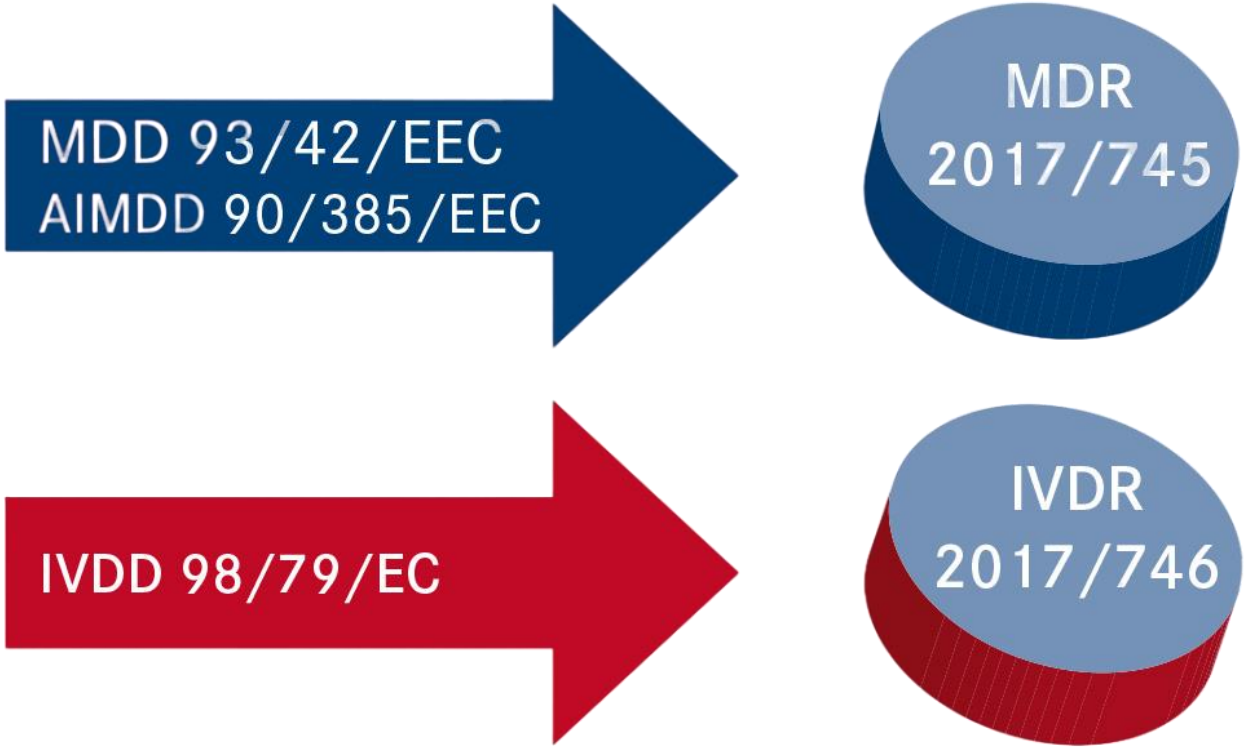


Time to market

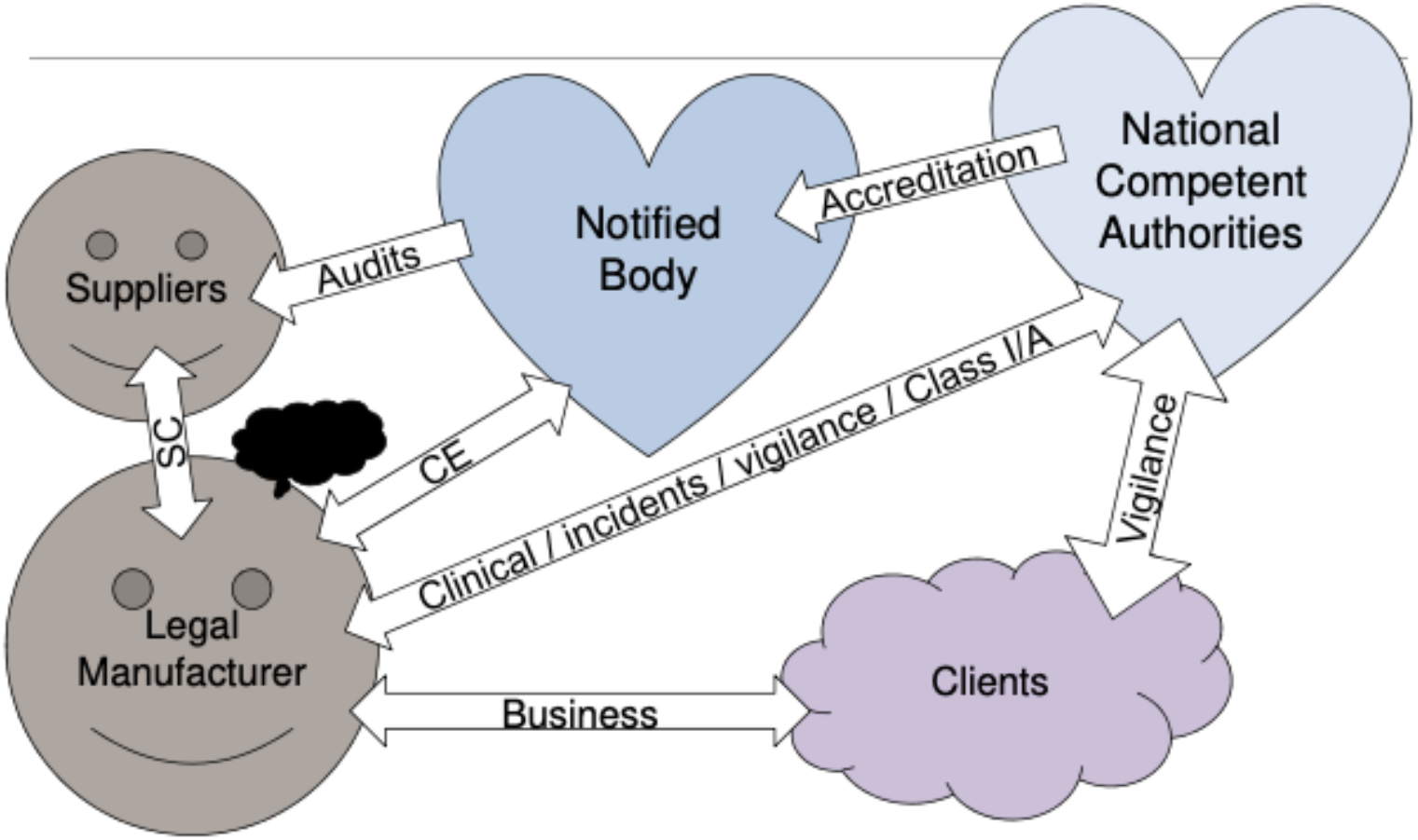
Price

Medical Devices and IVDs in EU

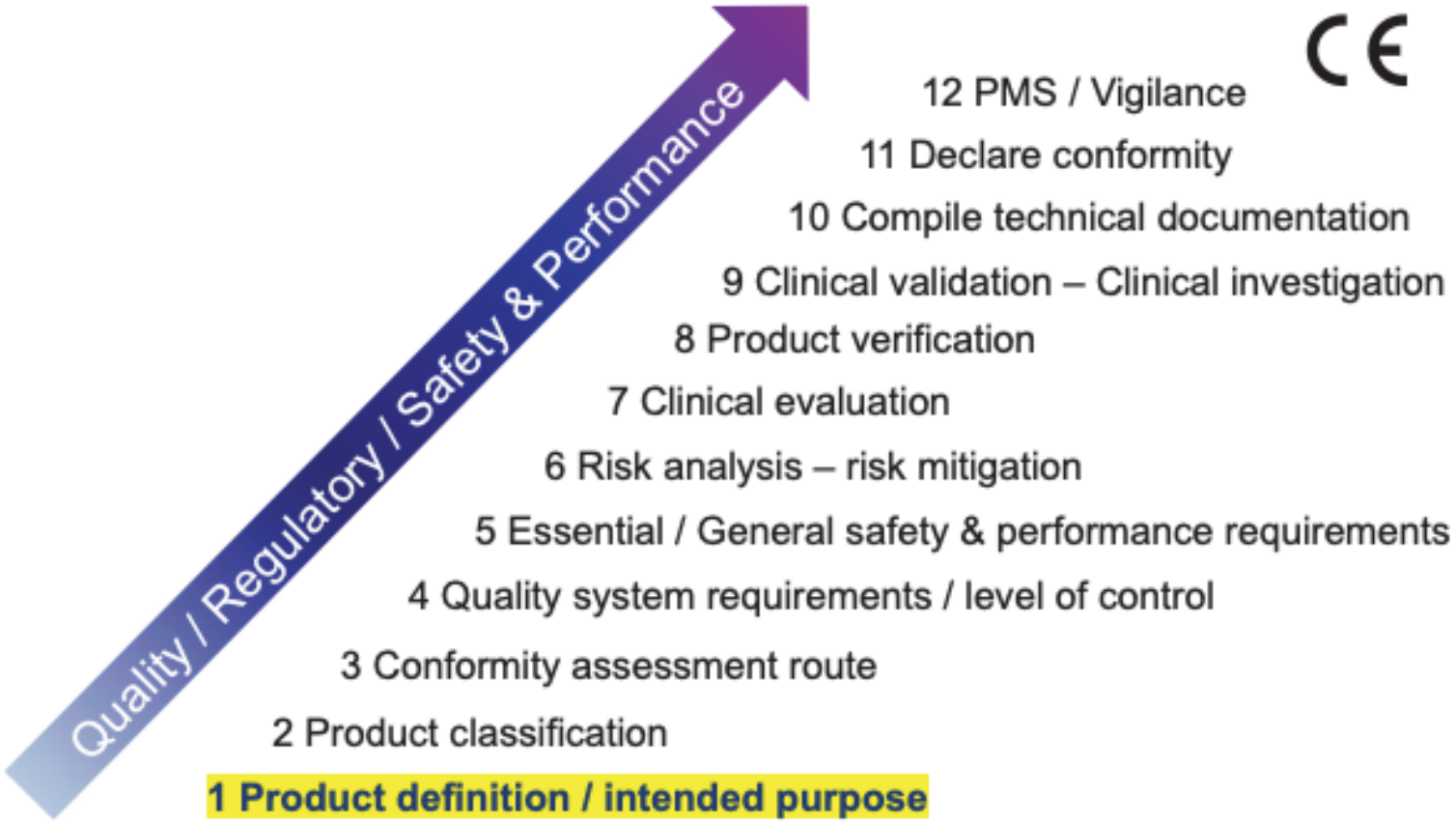
MEDICAL DEVICES AND IN VITRO DIAGNOSTICS IN EU



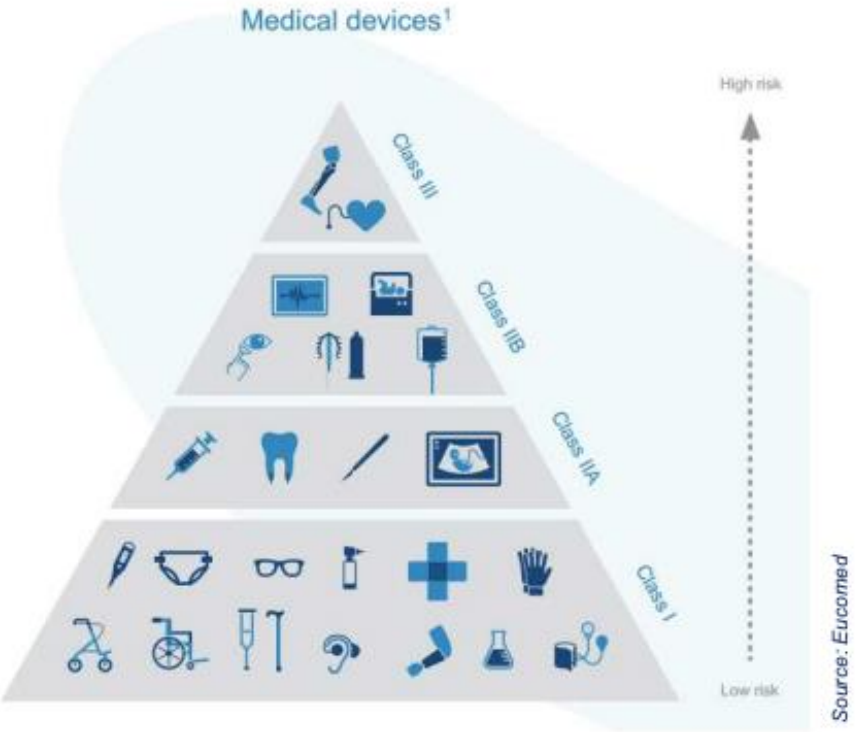
REGULATORY STAKEHOLDERS

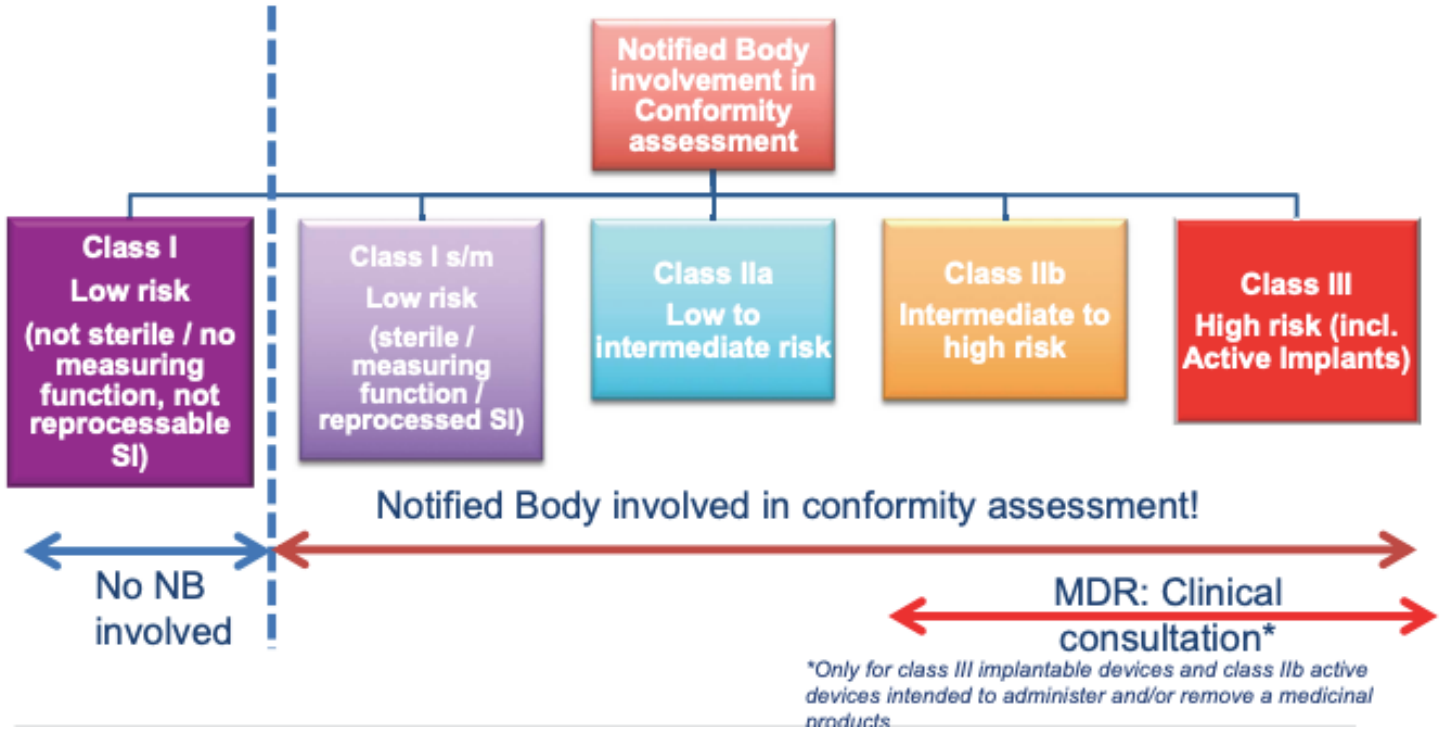


PATH TO MARKET



RISK BASED CLASSIFICATION



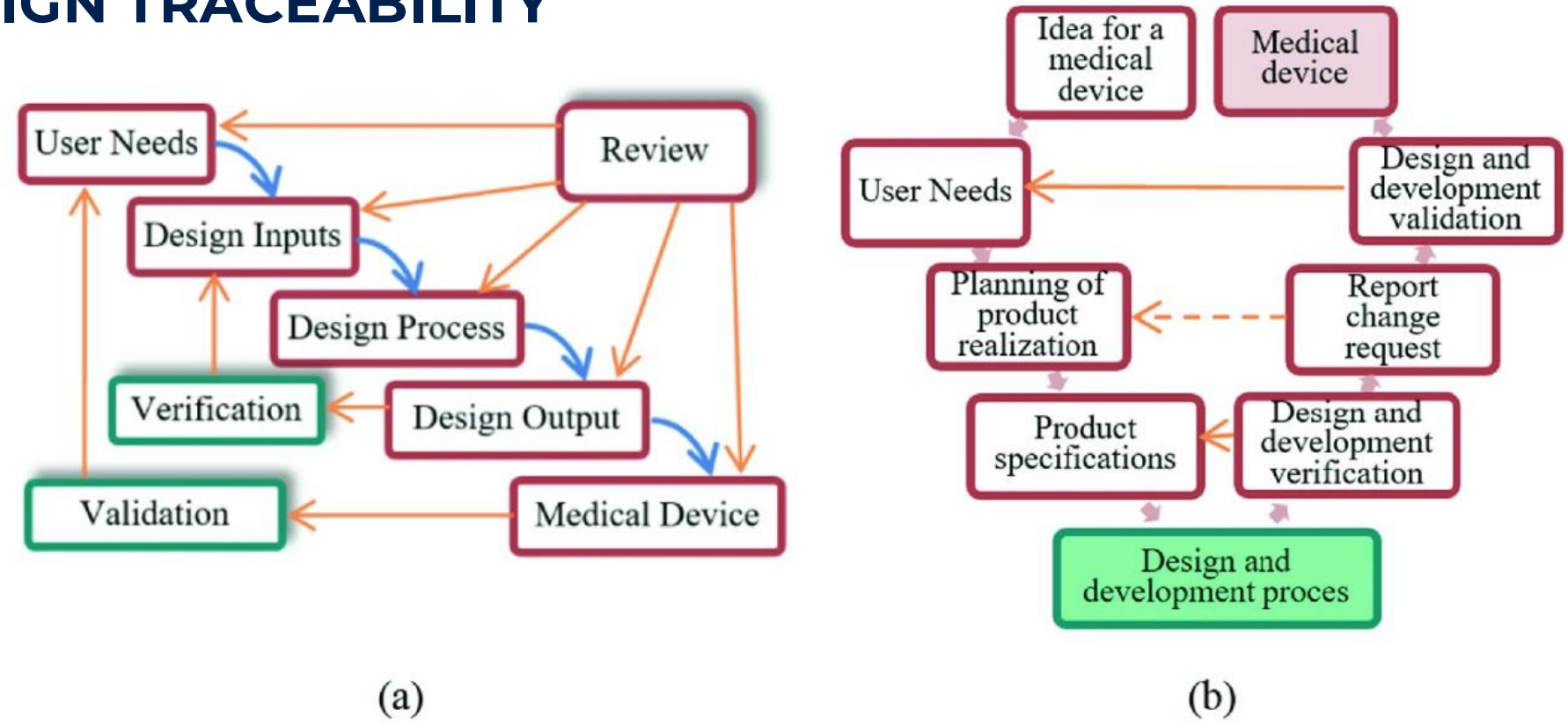


GENERAL SAFETY AND PERFORMANCE REQUIREMENTS

Requirement	Description	Applicable (yes/no)	Applied Standards (Annex Z)	Fulfilment of the requirements
	CHAPTER I. General Requirements			
1	Devices shall achieve the performance intended by their manufacturer and shall be designed and manufactured in such a way that, during normal conditions of use, they are suitable for their intended purpose. They shall be safe and effective and shall not compromise the clinical condition or the safety of patients, or the safety and health of users or, where applicable, other persons, provided that any risks which may be associated with their use constitute acceptable risks when weighed against the benefits to the patient and are compatible with a high level of protection of health and safety, taking into account the generally acknowledged state of the art.	Yes	EN ISO 14971 EN ISO 14155 ISO 14708-1 ISO 14708-3 EN 60601-1 EN 60601-1-2 EN 62304 EN ISO 13485	Hazard Analysis Report Risk Management Report System Verification Report Clinical Evidence Report EN 60601-1 Test Report EN 60601-1-2 Test Report
2	The requirement in this Annex to reduce risks as far as possible means the reduction of risks as far as possible without adversely affecting the benefit-risk ratio.	Yes	EN ISO 14971	Risk Management Report
…	…			
11.3	Devices labelled as sterile shall be processed, manufactured, packaged and, sterilised by means of appropriate, validated methods.	Yes	EN 556-1 EN ISO 11607-1 EN ISO 11607-2	Product label design Sterilisation validation plan Sterilisation validation report

- Other regulations
- Guidelines
- EU Standards
- State-of-the-art

DESIGN TRACEABILITY



Product design and development models: (a) Waterfall model and (b) V-model.

Ivanović, L., Rackov, M., Stojanović, B., Penčić, M., Čavić, M., Miladinović, S. (2022). Design and Development of Medical Devices. In: Lalic, B., Gracanin, D., Tasic, N., Simeunović, N. (eds) Proceedings on 18th International Conference on Industrial Systems – IS'20. IS 2020. Lecture Notes on Multidisciplinary Industrial Engineering. Springer, Cham. https://doi.org/10.1007/978-3-030-97947-8_6

RISK MANAGEMENT

NORME INTERNATIONALE

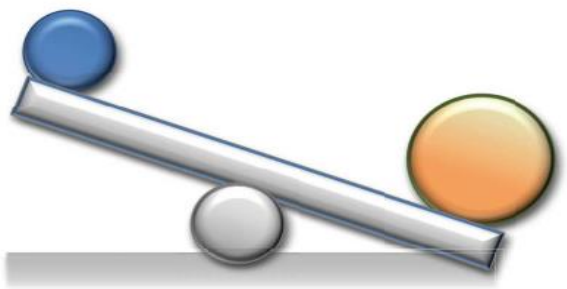
ISO 14971:2019

Edition 3
2019-12

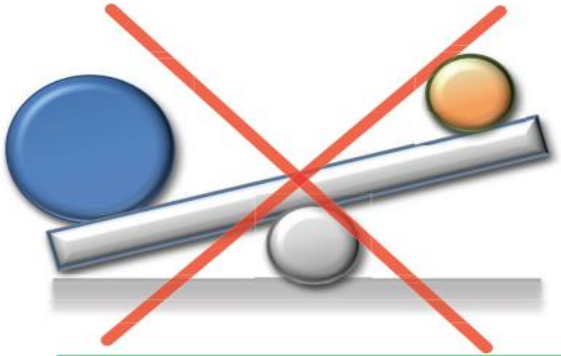
Dispositifs médicaux — Application de la gestion des risques aux dispositifs médicaux

ISO 14971:2019

Dispositifs médicaux — Application de la gestion des risques aux dispositifs médicaux

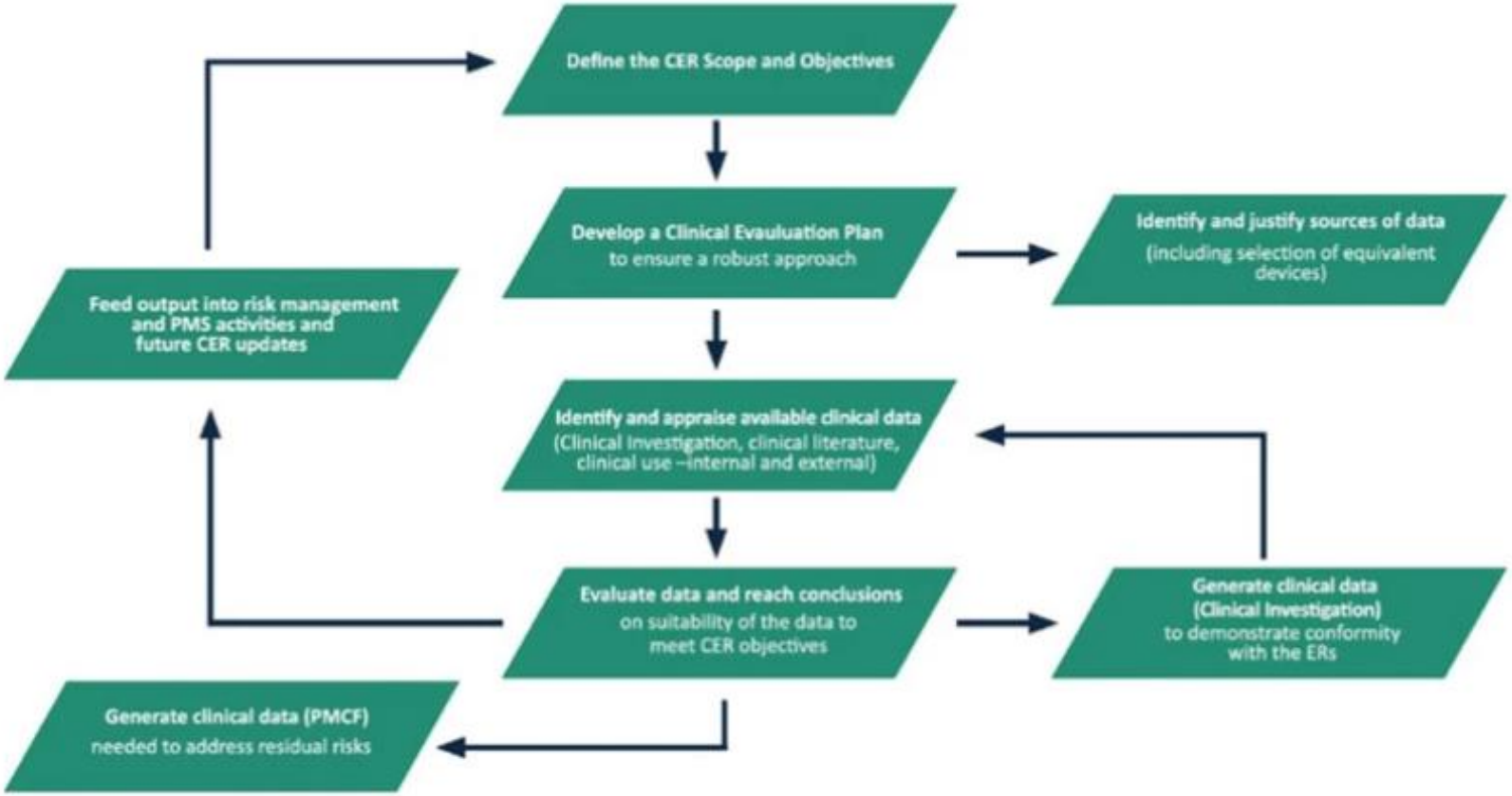


- Residual risk of device
- Risk of procedure



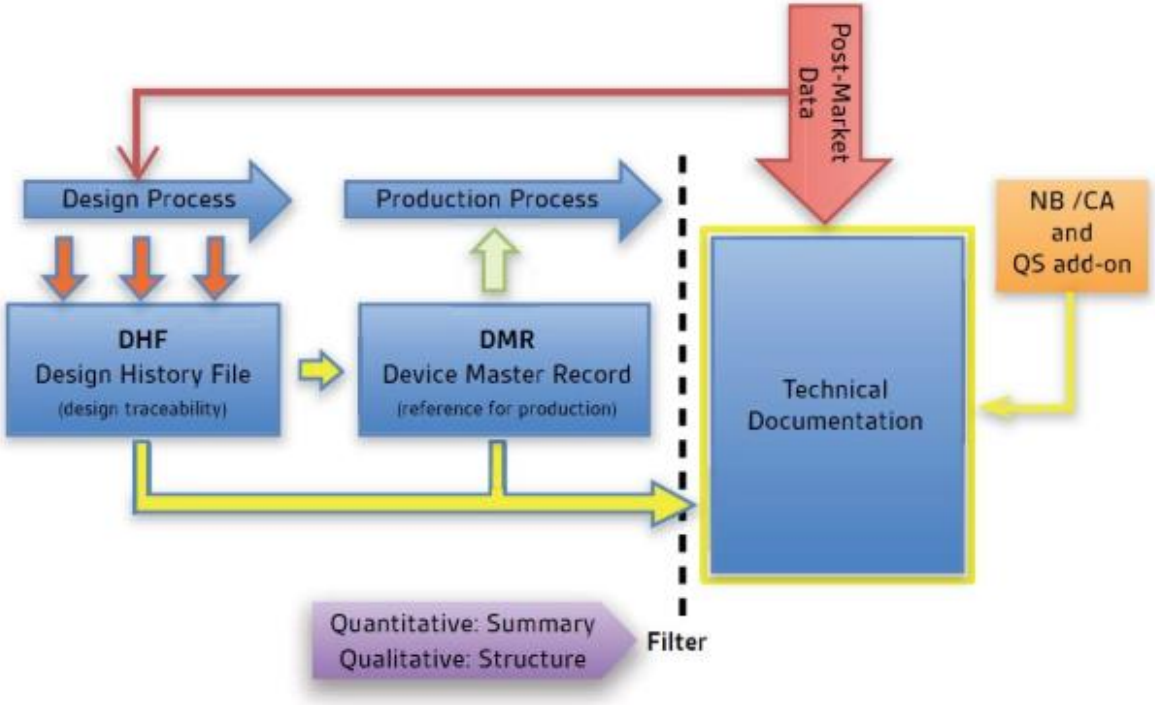
- Benefits for patient

CLINICAL EVALUATION AND CLINICAL INVESTIGATIONS

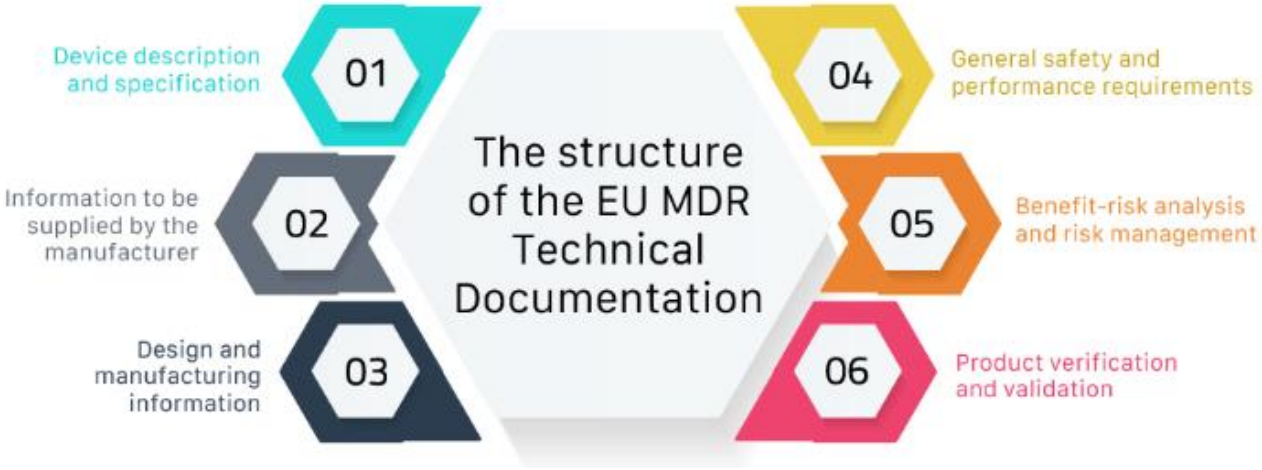


https://www.medicaldesignbriefs.com/component/content/article/33574-tackling-clinical-evaluation-head-on

TECHNICAL DOCUMENTATION

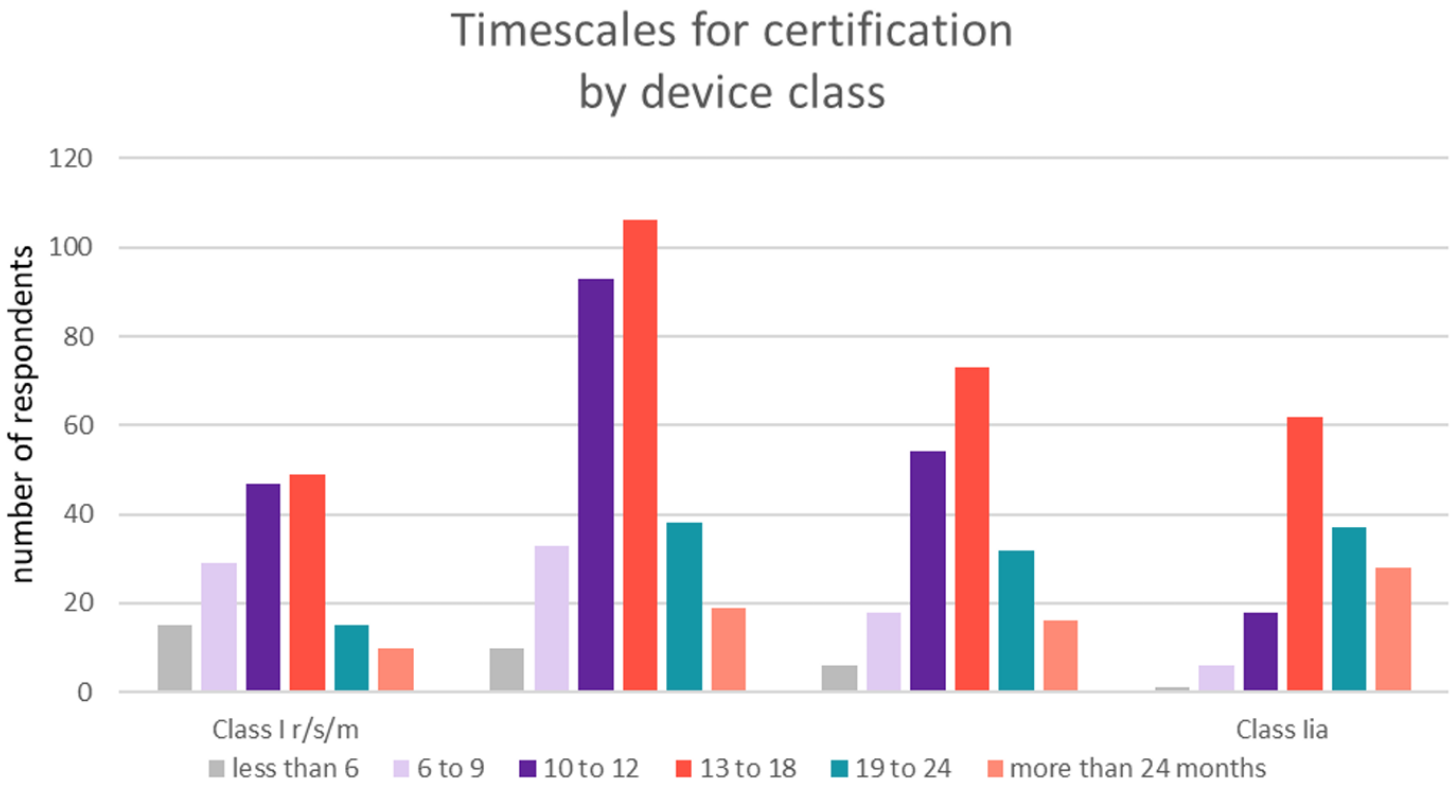


https://www.linkedin.com/pulse/mdr-technical-file-cenk-kili%C3%A7-kalkan/



https://advisera.com/13485academy/blog/2021/04/06/what-are-the-eu-mdr-technical-documentation-structure-and-requirements/

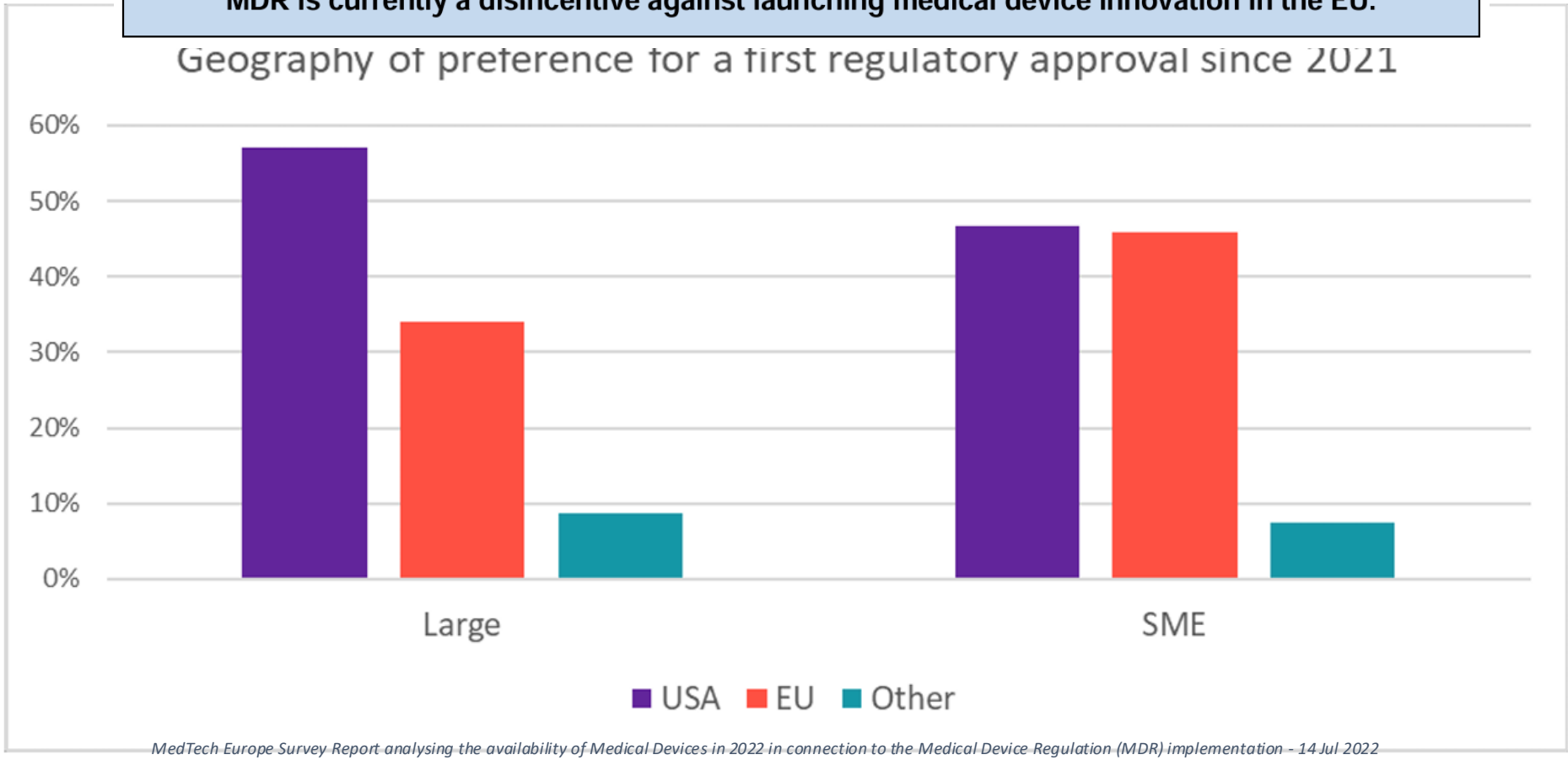
CONFORMITY ASSESSMENT TIMELINES



MedTech Europe Survey Report analysing the availability of Medical Devices in 2022 in connection to the Medical Device Regulation (MDR) implementation - 14 Jul 2022

IMPACT ON HEALTH INNOVATION IN EU

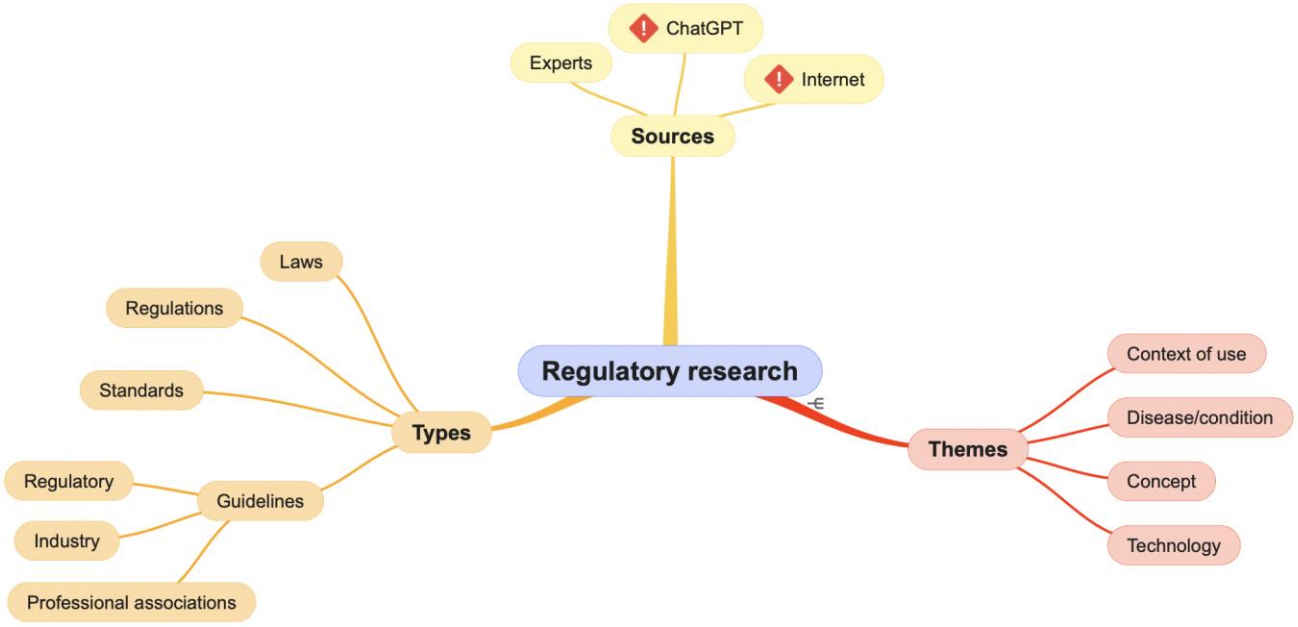
MDR is currently a disincentive against launching medical device innovation in the EU.

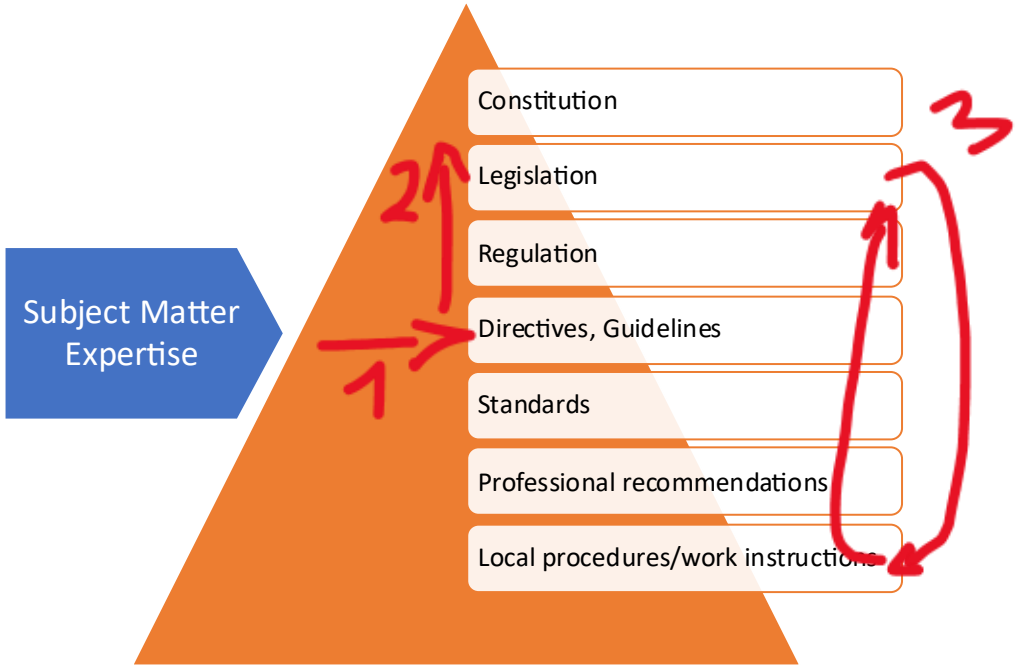


MedTech Europe Survey Report analysing the availability of Medical Devices in 2022 in connection to the Medical Device Regulation (MDR) implementation - 14 Jul 2022

Let’s get to it!

HOW TO ?


ChatGPT
Experts
Internet
Sources
Laws
Regulations
Standards
Types
Regulatory research
Themes
Context of use
Disease/condition
Concept
Technology
Regulatory
Guidelines
Industry
Professional associations


Subject Matter Expertise
Constitution
Legislation
Regulation
Directives, Guidelines
Standards
Professional recommendations
Local procedures/work instructions
1
2
3

YOUR TURN

For your project, for Belgium

- Query Google, prompt LLMs
- Any applicable laws, regulations, guidelines,…
  - Context of use, service
  - Disease, condition
  - Concept, technology, product
- Do they impact your solution? Your BMC?
  - Desirability? +/-
  - Feasibility? +/-
  - Viability? +/-

NEXT STEPS (Moodle will be updated in forthcoming days)

- Iterate on your concept using BMC and integrating financial and regulatory aspects
  - Feasible?
  - Viable?
- April 14th → Mind and Market (M&M) LLN: attend minimum 3 pitches
- April 15th 11:59pm → upload updated portfolio and presentation support in Moodle
- April 16th → Consultation #3
- After April 17th → present concept to user → feedback → final iterations
- May 7th → Session on pitching → come prepared based on M&M learnings
- May 12th → private defence
- May 13th → public pitch at closure ceremony

GEENS Pierre - MME, ieMBA

Project Leader – Health Innovation Classes
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